

# 09 — Business Metrics

All assumptions, rates, scenarios, and phase definitions in this document are sourced from [params.md](#). Refer to that document for any parameter clarification.

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## 1. Executive Summary

This document defines the complete metrics framework for tracking Health Hub's performance from pre-pilot internal testing through production-scale operations. The framework is designed for three audiences: (1) founders making day-to-day operational decisions, (2) investors evaluating traction and capital efficiency, and (3) operations managers monitoring service quality.

Health Hub operates as both a technology platform and an admin-assisted healthcare delivery operation (per [params.md](#) Section 1). This dual nature requires metrics that span traditional SaaS KPIs (MRR, churn, LTV:CAC) and healthcare-specific operational indicators (doctor utilization, consultation completion, callback conversion). The framework tracks 15 core business metrics and 5 operational metrics across all 4 phases and 21 scenario combinations.

### Key targets at steady-state production:

- LTV:CAC ratio > 3x (per [params.md](#) Section 12)
- Doctor utilization rate > 60% during operational hours (per [params.md](#) Section 12)
- Monthly active patients > 1,000 by end of Pilot 2
- Monthly burn rate sustainable for 6+ months at any checkpoint
- Consultation completion rate > 85%
- DAU/MAU ratio > 25% (indicating strong engagement)

## 2. Metric Definitions

### 2.1 Customer Acquisition Cost (CAC)

| Attribute             | Detail                                                                                                                                                                                                                          |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Total cost of acquiring one new paying user, including marketing spend, sales effort, onboarding cost, and partner activation cost                                                                                              |
| Formula               | $CAC = (\text{Total acquisition spend in period}) / (\text{New paying users acquired in period})$                                                                                                                               |
| Includes              | Digital marketing, referral incentives, partner clinic activation, community health worker outreach, admin staff time spent on onboarding                                                                                       |
| Excludes              | Product development costs, infrastructure, existing user retention spend                                                                                                                                                        |
| Measurement frequency | Monthly                                                                                                                                                                                                                         |
| Why it matters        | In East African markets where willingness to pay is constrained (\$28/year health expenditure per capita in Ethiopia per <a href="#">params.md</a> Section 9.1), CAC must remain extremely low to achieve viable unit economics |

### 2.2 Lifetime Value (LTV)

| Attribute             | Detail                                                                                                              |
|-----------------------|---------------------------------------------------------------------------------------------------------------------|
| Definition            | Total net revenue expected from a single user over their entire relationship with Health Hub                        |
| Formula               | $LTV = ARPU \times \text{Gross Margin \%} \times (1 / \text{Monthly Churn Rate})$                                   |
| Components            | Consultation fees, pharmacy commissions, diagnostics commissions, premium subscription fees, care coordination fees |
| Time horizon          | Modeled over 24-month expected user lifetime (conservative for emerging markets)                                    |
| Measurement frequency | Quarterly (requires sufficient cohort data)                                                                         |
| Why it matters        | LTV determines maximum justifiable CAC and signals whether the business model generates surplus value per user      |

### 2.3 LTV:CAC Ratio

| Attribute     | Detail                                                                                    |
|---------------|-------------------------------------------------------------------------------------------|
| Definition    | Ratio of user lifetime value to cost of acquiring that user                               |
| Formula       | $LTV:CAC = LTV / CAC$                                                                     |
| Target        | > 3x by Production (per <a href="#">params.md</a> Section 12)                             |
| Healthy range | 3x-5x is sustainable; < 2x is unsustainable; > 5x may indicate under-investment in growth |

|                       |                                                                                                                            |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Measurement frequency | Quarterly                                                                                                                  |
| Why it matters        | The single most important unit economics metric. Below 3x, the company burns capital faster than it creates value per user |

2.4 Churn Rate

| Attribute             | Detail                                                                                                                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Percentage of active users who stop using the platform in a given period                                                                                                   |
| Formula               | Monthly Churn = (Users lost in month) / (Users at start of month) x 100                                                                                                    |
| User types tracked    | Patient churn (primary), facility churn (B2B), doctor churn (supply-side)                                                                                                  |
| Measurement frequency | Monthly                                                                                                                                                                    |
| Why it matters        | Healthcare platforms in emerging markets face high churn from connectivity issues, alternative care-seeking behavior, and price sensitivity. Churn directly compresses LTV |

2.5 Average Revenue Per User (ARPU)

| Attribute             | Detail                                                                                                      |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| Definition            | Average monthly revenue generated per active user                                                           |
| Formula               | ARPU = Total monthly revenue / Monthly active users                                                         |
| Segmented             | Patient ARPU, facility ARPU, blended ARPU                                                                   |
| Measurement frequency | Monthly                                                                                                     |
| Why it matters        | Tracks monetization efficiency per user; critical for validating pricing corridors (params.md Section 11.2) |

2.6 Monthly Recurring Revenue (MRR)

| Attribute             | Detail                                                                                                                                        |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Predictable revenue received every month from subscriptions and recurring platform fees                                                       |
| Formula               | MRR = (Number of subscribers x subscription price) + (Number of facilities x platform fee)                                                    |
| Components            | Premium patient subscriptions (ETB 200-400/mo or KES 400-800/mo) + facility platform fees (ETB 2,000-5,000/mo or KES 5,000-10,000/mo)         |
| Measurement frequency | Monthly                                                                                                                                       |
| Why it matters        | MRR provides revenue predictability. Until MRR exceeds fixed costs, the company depends on transactional revenue which is inherently volatile |

2.7 Annual Recurring Revenue (ARR)

| Attribute             | Detail                                                                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
| Definition            | MRR annualized                                                                                                             |
| Formula               | ARR = MRR x 12                                                                                                             |
| Measurement frequency | Monthly (calculated)                                                                                                       |
| Why it matters        | Standard SaaS valuation metric. Investors benchmark Health Hub's ARR against comparable East African health tech companies |

2.8 Burn Rate

| Attribute             | Detail                                                                                                                                                                                                           |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Net cash consumed per month after revenue                                                                                                                                                                        |
| Formula               | Burn Rate = Total monthly outflow - Total monthly revenue                                                                                                                                                        |
| Components            | Team cost (\$14,830/mo per params.md Section 3.2) + ops staff (\$0-\$6,542/mo by phase per params.md Section 4.2) + infra (\$65-\$800/mo midpoint by phase per params.md Section 7.4) + one-time costs amortized |
| Measurement frequency | Monthly                                                                                                                                                                                                          |
| Why it matters        | Burn rate divided into remaining cash equals runway. For a self-funded company (\$1), burn rate is the primary survival constraint                                                                               |

2.9 Runway

| Attribute             | Detail                                                       |
|-----------------------|--------------------------------------------------------------|
| Definition            | Number of months of operation remaining at current burn rate |
| Formula               | Runway = Cash on hand / Monthly burn rate                    |
| Critical threshold    | < 2 months triggers contingency actions                      |
| Measurement frequency | Monthly                                                      |

|                |                                                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Why it matters | The company dies when runway reaches zero. Per params.md Section 12, monthly burn rate must be sustainable for 6+ months at any point |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------|

2.10 Gross Margin

| Attribute             | Detail                                                                                                                                                     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Revenue minus direct cost of delivering the service, expressed as a percentage                                                                             |
| Formula               | Gross Margin = (Revenue - COGS) / Revenue x 100                                                                                                            |
| COGS includes         | Doctor salaries for employed doctors, video infrastructure (Daily.co), SMS/notification costs, payment processing fees                                     |
| COGS excludes         | Team salaries (below the line), marketing, admin overhead                                                                                                  |
| Measurement frequency | Monthly                                                                                                                                                    |
| Why it matters        | Gross margin must be positive before the business model is viable. Low gross margin in healthcare indicates mispriced services or excessive delivery costs |

2.11 Net Margin

| Attribute             | Detail                                                                                                                                            |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Total revenue minus all costs (COGS + operating expenses + team + infra), expressed as a percentage                                               |
| Formula               | Net Margin = (Revenue - All Costs) / Revenue x 100                                                                                                |
| Measurement frequency | Monthly                                                                                                                                           |
| Why it matters        | Net margin positive = the company is self-sustaining. Expected to be deeply negative during Pilot phases and approach breakeven during Production |

2.12 DAU/MAU Ratio (Stickiness)

| Attribute             | Detail                                                                                                                                                                                             |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Ratio of daily active users to monthly active users, measuring how often users return                                                                                                              |
| Formula               | DAU/MAU = Average daily active users / Monthly active users                                                                                                                                        |
| Target                | > 25% by Production                                                                                                                                                                                |
| Measurement frequency | Daily (rolling 30-day MAU)                                                                                                                                                                         |
| Why it matters        | Healthcare apps typically have lower DAU/MAU than social apps but higher than banking apps. A ratio below 15% suggests the platform is used only for acute episodes, not as a primary care gateway |

2.13 Consultation Completion Rate

| Attribute             | Detail                                                                                                                                                                                                       |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Percentage of initiated consultations that reach a completed state (diagnosis + next-step issued)                                                                                                            |
| Formula               | Completion Rate = Completed consultations / Initiated consultations x 100                                                                                                                                    |
| Excludes              | Cancelled before doctor assignment, test/demo consultations                                                                                                                                                  |
| Target                | > 85% by Production                                                                                                                                                                                          |
| Measurement frequency | Weekly                                                                                                                                                                                                       |
| Why it matters        | Incomplete consultations represent failed service delivery. They generate no revenue, damage trust, and increase churn. Low completion rate signals tech issues (video drops), staffing gaps, or UX problems |

2.14 Doctor Utilization Rate

| Attribute             | Detail                                                                                                                                                                                                         |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Percentage of available doctor hours that are spent in active consultations                                                                                                                                    |
| Formula               | Utilization = (Hours in consultations) / (Total scheduled doctor hours) x 100                                                                                                                                  |
| Target                | > 60% during operational hours (per params.md Section 12)                                                                                                                                                      |
| Measurement frequency | Daily                                                                                                                                                                                                          |
| Why it matters        | Employed doctors (INR 60,000/mo per params.md Section 4.1) are a fixed cost. Low utilization means overstaffing; high utilization (>85%) means patient wait times are unacceptable. 60-75% is the healthy band |

2.15 Net Promoter Score (NPS)

| Attribute             | Detail                                                                                                                                                                                                                 |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition            | Likelihood that a user would recommend Health Hub to others, measured on a 0-10 scale                                                                                                                                  |
| Formula               | NPS = % Promoters (9-10) - % Detractors (0-6)                                                                                                                                                                          |
| Collection method     | In-app survey after consultation completion; quarterly email survey for inactive users                                                                                                                                 |
| Target                | > 40 by Production (good); > 60 is excellent                                                                                                                                                                           |
| Measurement frequency | Quarterly                                                                                                                                                                                                              |
| Why it matters        | In markets where word-of-mouth drives adoption more than digital marketing, NPS is a leading indicator of organic growth. Ethiopian and Kenyan healthcare decisions are heavily influenced by personal recommendations |

### 3. Metrics by Phase

#### 3.1 Pre-Pilot Phase (Months 1-6, 100-200 internal users)

| Metric                       | Target Value                        | Rationale                                                                        |
|------------------------------|-------------------------------------|----------------------------------------------------------------------------------|
| CAC                          | Not measured (internal users)       | No external acquisition; users are team members, advisors, partner staff         |
| LTV                          | Not measured                        | No revenue; testing phase                                                        |
| LTV:CAC                      | N/A                                 | No revenue                                                                       |
| Churn Rate                   | < 20% monthly                       | Internal users should stay engaged; high churn signals UX problems               |
| ARPU                         | \$0                                 | No monetization                                                                  |
| MRR                          | \$0                                 | No subscriptions active                                                          |
| ARR                          | \$0                                 | No subscriptions active                                                          |
| Burn Rate                    | \$14,830-\$14,980/mo (team + infra) | Team \$14,830 + infra \$65-\$150 (per params.md Sections 3.2, 7.4); no ops staff |
| Runway                       | Depends on starting capital         | Must be > 6 months                                                               |
| Gross Margin                 | N/A                                 | No revenue                                                                       |
| Net Margin                   | N/A                                 | No revenue                                                                       |
| DAU/MAU                      | > 30%                               | Internal testers should be active daily; signals platform readiness              |
| Consultation Completion Rate | > 70%                               | Lower target acceptable; focus is on identifying failure modes                   |
| Doctor Utilization           | N/A                                 | No employed doctors yet                                                          |
| NPS                          | > 30 (internal benchmark)           | Team members providing honest feedback                                           |

#### 3.2 Pilot 1 Phase (Months 5-9, up to 1,000 users)

| Metric                       | Target Value           | Rationale                                                                                                                            |
|------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| CAC                          | < \$5 per user         | Low-cost acquisition through partner clinics, community outreach; subsidized onboarding                                              |
| LTV                          | \$8-\$15 (estimated)   | Based on 6-12 month retention at subsidized pricing (ETB 150-300 per GP consult)                                                     |
| LTV:CAC                      | > 1.5x                 | Not yet at 3x target; proving directional viability                                                                                  |
| Churn Rate                   | < 15% monthly          | Early users are curated; lower churn expected from engaged pilot cohort                                                              |
| ARPU                         | \$0.50-\$2.00/mo       | Subsidized/symbolic pricing; \$0-\$500/mo total revenue across user base (per params.md Section 11.3)                                |
| MRR                          | \$0-\$100              | Subscriptions not yet active; any recurring revenue is a bonus                                                                       |
| ARR                          | \$0-\$1,200            | Annualized from minimal MRR                                                                                                          |
| Burn Rate                    | \$17,312-\$17,832/mo   | Team \$14,830 + ops staff \$2,482 (per params.md Section 4.2) + infra \$270 midpoint (per params.md Section 7.4) - revenue \$0-\$500 |
| Runway                       | Must remain > 6 months | Critical checkpoint before committing to Pilot 2                                                                                     |
| Gross Margin                 | -50% to 0%             | Revenue likely does not cover direct service delivery costs at subsidized rates                                                      |
| Net Margin                   | < -500%                | Deeply negative; expected and acceptable during pilot                                                                                |
| DAU/MAU                      | > 20%                  | Real users; healthcare usage is episodic. 20%+ signals meaningful engagement                                                         |
| Consultation Completion Rate | > 75%                  | Improving from Pre-Pilot; real patient consultations must complete reliably                                                          |
| Doctor Utilization           | 30-50%                 | 2 employed doctors (per params.md Section 4.2); demand still building                                                                |
| NPS                          | > 35                   | Early adopters tend to be more forgiving; baseline for improvement tracking                                                          |

3.3 Pilot 2 Phase (Months 8-18, 5,000-10,000 users)

| Metric                       | Target Value           | Rationale                                                                                                                                |
|------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| CAC                          | < \$3 per user         | Economies of scale; word-of-mouth reducing paid acquisition share                                                                        |
| LTV                          | \$15-\$35              | Pricing at 50% of target (per params.md Section 11.3); longer retention emerging                                                         |
| LTV:CAC                      | > 2.5x                 | Approaching the 3x target; unit economics becoming clear                                                                                 |
| Churn Rate                   | < 10% monthly          | Users seeing real value; sticky workflows (prescriptions, lab results) reducing churn                                                    |
| ARPU                         | \$2-\$5/mo             | 50% of target pricing active; specialist consults available                                                                              |
| MRR                          | \$200-\$1,500          | Early premium subscriptions + some facility platform fees                                                                                |
| ARR                          | \$2,400-\$18,000       | Meaningful ARR for investor narrative                                                                                                    |
| Burn Rate                    | \$19,191-\$20,091/mo   | Team \$14,830 + ops staff \$4,361 (per params.md Section 4.2) + infra \$460 midpoint (per params.md Section 7.4) - revenue \$500-\$3,000 |
| Runway                       | Must remain > 6 months | Investor capital may have arrived (S2, S3); extends runway significantly                                                                 |
| Gross Margin                 | 10-30%                 | Revenue beginning to cover direct delivery costs                                                                                         |
| Net Margin                   | -200% to -100%         | Still negative but improving rapidly                                                                                                     |
| DAU/MAU                      | > 22%                  | Growing user base with episodic but recurring healthcare needs                                                                           |
| Consultation Completion Rate | > 80%                  | System reliability improving; operational SOPs in place                                                                                  |
| Doctor Utilization           | 50-65%                 | 4 doctors (per params.md Section 4.2); approaching target utilization                                                                    |
| NPS                          | > 40                   | Service quality must be consistently good to sustain growth                                                                              |

3.4 Production Phase (Months 16-34, full scale)

| Metric                       | Target Value                            | Rationale                                                                                                                                   |
|------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| CAC                          | < \$2 per user                          | Strong word-of-mouth; organic growth dominant; partner network driving referrals                                                            |
| LTV                          | \$30-\$60                               | Full pricing active; 18-24 month expected lifetime; multiple revenue streams per user                                                       |
| LTV:CAC                      | > 3x (target per params.md Section 12)  | Unit economics proven; growth investment justified                                                                                          |
| Churn Rate                   | < 8% monthly                            | Platform deeply integrated into user healthcare journey                                                                                     |
| ARPU                         | \$5-\$15/mo                             | Full pricing across GP, specialist, pharmacy, diagnostics; premium subscriptions                                                            |
| MRR                          | \$3,000-\$15,000                        | Per params.md Section 11.3; subscriptions + facility fees providing stable base                                                             |
| ARR                          | \$36,000-\$180,000                      | Meaningful ARR; basis for Series A narrative                                                                                                |
| Burn Rate                    | \$18,830-\$22,172/mo declining          | Team \$14,830 + ops staff \$6,542 (per params.md Section 4.2) + infra \$800 midpoint (per params.md Section 7.4) - revenue \$3,000-\$15,000 |
| Runway                       | > 12 months or self-sustaining          | Approaching breakeven; burn rate declining as revenue scales                                                                                |
| Gross Margin                 | 40-60%                                  | Technology platform economics; marginal cost of additional consultation is low                                                              |
| Net Margin                   | -50% to +10%                            | Approaching breakeven; best-case paths achieve positive net margin                                                                          |
| DAU/MAU                      | > 25%                                   | Target engagement level; users treating Health Hub as primary care gateway                                                                  |
| Consultation Completion Rate | > 85%                                   | Mature operational processes; reliable technology; trained doctor pool                                                                      |
| Doctor Utilization           | > 60% (target per params.md Section 12) | 6 doctors (per params.md Section 4.2); scheduling optimized                                                                                 |
| NPS                          | > 50                                    | Market-leading satisfaction; drives organic growth                                                                                          |

3.5 Phase Progression Summary

| Metric          | Pre-Pilot | Pilot 1    | Pilot 2       | Production       |
|-----------------|-----------|------------|---------------|------------------|
| CAC             | N/A       | < \$5      | < \$3         | < \$2            |
| LTV             | N/A       | \$8-\$15   | \$15-\$35     | \$30-\$60        |
| LTV:CAC         | N/A       | > 1.5x     | > 2.5x        | > 3x             |
| Churn (monthly) | < 20%     | < 15%      | < 10%         | < 8%             |
| ARPU            | \$0       | \$0.50-\$2 | \$2-\$5       | \$5-\$15         |
| MRR             | \$0       | \$0-\$100  | \$200-\$1,500 | \$3,000-\$15,000 |

|                    |           |             |                  |                     |
|--------------------|-----------|-------------|------------------|---------------------|
| ARR                | \$0       | \$0-\$1,200 | \$2,400-\$18,000 | \$36,000-\$180,000  |
| Burn Rate/mo       | ~\$14,900 | ~\$17,600   | ~\$19,600        | ~\$20,500 declining |
| Gross Margin       | N/A       | -50% to 0%  | 10-30%           | 40-60%              |
| Net Margin         | N/A       | < -500%     | -200% to -100%   | -50% to +10%        |
| DAU/MAU            | > 30%     | > 20%       | > 22%            | > 25%               |
| Consult Completion | > 70%     | > 75%       | > 80%            | > 85%               |
| Doctor Utilization | N/A       | 30-50%      | 50-65%           | > 60%               |
| NPS                | > 30      | > 35        | > 40             | > 50                |

## 4. Metrics by Scenario

### 4.1 How Scenarios Affect Metrics

The 21 combinations (per params.md Section 15) affect metrics through three primary channels:

- Speed of user acquisition** — Higher investment levels (L2, L3) fund more marketing and partner activation, reducing CAC but increasing total spend
- Ops staff ramp** — Higher levels allow faster hiring of doctors, admin, and tech support, improving consultation completion and utilization
- Runway pressure** — Self-funded paths (S1) face constant pressure to reduce burn, which constrains growth investment and delays LTV:CAC maturity

### 4.2 Summary Table: Key Metrics Across All 21 Combinations at End of Pilot 2

| #  | Code     | CAC     | LTV       | LTV:CAC | Churn/mo | ARPU   | MRR     | Burn Rate/mo | Runway Risk |
|----|----------|---------|-----------|---------|----------|--------|---------|--------------|-------------|
| 1  | S1-L1    | \$6-\$8 | \$8-\$12  | 1.2x    | 18%      | \$1.00 | \$50    | \$8,700      | HIGH        |
| 2  | S1-L2    | \$4-\$5 | \$12-\$20 | 2.5x    | 12%      | \$2.50 | \$400   | \$17,600     | MEDIUM      |
| 3  | S1-L3    | \$3-\$4 | \$18-\$30 | 5.0x    | 9%       | \$4.00 | \$1,200 | \$28,000     | MEDIUM      |
| 4  | S2-O1-I1 | \$5-\$7 | \$10-\$15 | 1.5x    | 15%      | \$1.50 | \$100   | \$9,500      | MEDIUM      |
| 5  | S2-O1-I2 | \$4-\$5 | \$12-\$22 | 3.0x    | 11%      | \$2.50 | \$400   | \$17,600     | LOW         |
| 6  | S2-O1-I3 | \$3-\$4 | \$15-\$28 | 4.5x    | 9%       | \$3.50 | \$900   | \$26,000     | LOW         |
| 7  | S2-O2-I1 | \$4-\$6 | \$12-\$18 | 2.0x    | 13%      | \$2.00 | \$250   | \$12,500     | MEDIUM      |
| 8  | S2-O2-I2 | \$3-\$4 | \$15-\$25 | 3.5x    | 10%      | \$3.50 | \$800   | \$19,600     | LOW         |
| 9  | S2-O2-I3 | \$2-\$3 | \$18-\$32 | 5.5x    | 8%       | \$4.50 | \$1,400 | \$28,500     | LOW         |
| 10 | S2-O3-I1 | \$3-\$5 | \$15-\$22 | 3.0x    | 11%      | \$3.00 | \$500   | \$18,000     | MEDIUM      |
| 11 | S2-O3-I2 | \$2-\$3 | \$18-\$30 | 5.0x    | 9%       | \$4.00 | \$1,100 | \$24,000     | LOW         |
| 12 | S2-O3-I3 | \$2-\$3 | \$22-\$35 | 6.0x    | 7%       | \$5.00 | \$1,800 | \$32,000     | LOW         |
| 13 | S3-O1-I1 | \$5-\$7 | \$10-\$15 | 1.5x    | 15%      | \$1.50 | \$100   | \$9,500      | HIGH        |
| 14 | S3-O1-I2 | \$3-\$5 | \$14-\$24 | 3.5x    | 10%      | \$3.00 | \$500   | \$18,000     | LOW         |
| 15 | S3-O1-I3 | \$2-\$4 | \$18-\$30 | 5.0x    | 8%       | \$4.00 | \$1,200 | \$27,000     | LOW         |
| 16 | S3-O2-I1 | \$4-\$6 | \$12-\$18 | 2.0x    | 13%      | \$2.00 | \$250   | \$12,500     | MEDIUM      |
| 17 | S3-O2-I2 | \$3-\$4 | \$16-\$28 | 4.0x    | 9%       | \$3.50 | \$900   | \$20,000     | LOW         |
| 18 | S3-O2-I3 | \$2-\$3 | \$20-\$33 | 5.5x    | 7%       | \$4.50 | \$1,500 | \$29,000     | LOW         |
| 19 | S3-O3-I1 | \$3-\$5 | \$15-\$22 | 3.0x    | 11%      | \$3.00 | \$500   | \$18,000     | MEDIUM      |
| 20 | S3-O3-I2 | \$2-\$3 | \$20-\$32 | 5.5x    | 8%       | \$4.50 | \$1,300 | \$25,000     | LOW         |
| 21 | S3-O3-I3 | \$1-\$2 | \$25-\$38 | 8.0x    | 6%       | \$5.50 | \$2,000 | \$34,000     | LOW         |

Notes on the table:

- S1-L1 burn rate reflects 40% of ideal spending level (per params.md Section 14.2), hence lower absolute burn but also lower service quality and slower growth
- Runway risk is HIGH when remaining runway at end of Pilot 2 is < 4 months; MEDIUM when 4-8 months; LOW when > 8 months or investor-backed
- LTV:CAC ratios above 5x in heavily funded paths reflect lower CAC from economies of scale, not necessarily higher absolute LTV

### 4.3 Scenario Group Patterns

**S1 (Self-Funded) — 3 combinations:**

- CAC remains high due to limited marketing budget
- LTV growth is constrained by slower user acquisition (smaller network effects)
- Runway is the binding constraint; metrics take a back seat to survival
- S1-L1 is the highest-risk path; S1-L3 requires significant founder capital but produces the best self-funded metrics

**S2 (Investor During Transition) — 9 combinations:**

- Investor capital arrives at the Pilot 1-to-Pilot 2 transition, providing a runway buffer

- Combinations where investor level >= O2 (e.g., S2-O1-I2, S2-O2-I2) represent the best capital-efficient paths
- S2-O2-I2 is the baseline recommended path: balanced own spending with ideal investor matching

S3 (Investor After Pilot 1) — 9 combinations:

- Earlier investor arrival accelerates all metrics but requires giving up equity sooner (per params.md Section 8.3)
- S3-O3-I3 produces the best absolute metrics (LTV:CAC 8x, highest ARPU) but requires the most capital and carries the highest absolute burn
- S3-O1-I1 is essentially a "small money early" path that provides modest improvement over S1

## 5. Operational Metrics

### 5.1 Callback Conversion Rate

| Attribute       | Detail                                                                                                                                                                                        |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition      | Percentage of admin callbacks that result in a booked consultation or completed service                                                                                                       |
| Formula         | Conversion = (Callbacks resulting in booked service) / (Total callbacks made) x 100                                                                                                           |
| Target by phase | Pre-Pilot: N/A; Pilot 1: > 25%; Pilot 2: > 35%; Production: > 45%                                                                                                                             |
| Why it matters  | Health Hub's admin-assisted model (per params.md Section 1) depends on callbacks converting to revenue. Low conversion suggests poor targeting, wrong timing, or misaligned service offerings |

### 5.2 Average Response Time

| Attribute       | Detail                                                                                                                                                                                       |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition      | Time elapsed between a patient initiating a consultation request and receiving a response (doctor assignment or callback acknowledgment)                                                     |
| Measurement     | Median response time (not mean, to avoid outlier distortion)                                                                                                                                 |
| Target by phase | Pilot 1: < 30 minutes; Pilot 2: < 15 minutes; Production: < 10 minutes                                                                                                                       |
| Why it matters  | Response time is the primary driver of patient satisfaction in telehealth. In East African markets where patients are accustomed to long clinic waits, fast response is a key differentiator |

### 5.3 Doctor Coverage Hours

| Attribute       | Detail                                                                                                                 |
|-----------------|------------------------------------------------------------------------------------------------------------------------|
| Definition      | Percentage of target operating hours (e.g., 16 hours/day for 2-shift coverage) that have at least one doctor available |
| Formula         | Coverage = (Hours with >= 1 doctor online) / (Target operating hours) x 100                                            |
| Target by phase | Pilot 1: > 80% (2 doctors, 2 shifts per params.md Section 4.4); Pilot 2: > 90%; Production: > 98% (near-24/7)          |
| Why it matters  | Coverage gaps mean patients encounter "no doctor available" — the fastest way to lose trust and drive churn            |

### 5.4 Admin Resolution Time

| Attribute       | Detail                                                                                                                                   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Definition      | Average time for admin ops staff to resolve a patient issue (prescription fulfillment, lab booking, partner coordination, billing query) |
| Measurement     | Median time from ticket creation to resolution                                                                                           |
| Target by phase | Pilot 1: < 4 hours; Pilot 2: < 2 hours; Production: < 1 hour                                                                             |
| Why it matters  | Admin staff are a direct cost (\$217/mo per params.md Section 4.1); slow resolution means either understaffing or poor SOPs              |

### 5.5 Support Ticket Volume

| Attribute       | Detail                                                                                                                                                  |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition      | Number of support tickets created per 100 active users per month                                                                                        |
| Formula         | Volume = (Total tickets in month) / (MAU / 100)                                                                                                         |
| Target by phase | Pilot 1: < 30 per 100 users; Pilot 2: < 20 per 100 users; Production: < 12 per 100 users                                                                |
| Why it matters  | High ticket volume relative to user base signals UX problems, bugs, or confusing workflows. Declining ticket volume per user confirms platform maturity |

### 5.6 Operational Metrics Summary Table

| Metric                       | Pilot 1  | Pilot 2  | Production |
|------------------------------|----------|----------|------------|
| Callback conversion rate     | > 25%    | > 35%    | > 45%      |
| Median response time         | < 30 min | < 15 min | < 10 min   |
| Doctor coverage hours        | > 80%    | > 90%    | > 98%      |
| Median admin resolution time | < 4 hrs  | < 2 hrs  | < 1 hr     |

|                             |      |      |      |
|-----------------------------|------|------|------|
| Support tickets per 100 MAU | < 30 | < 20 | < 12 |
|-----------------------------|------|------|------|

## 6. Dashboard Design

### 6.1 Design Principles

The metrics dashboard serves three distinct audiences with different update frequencies and depth requirements:

| Audience            | Update Frequency  | Primary Concern                                                           | View Type                      |
|---------------------|-------------------|---------------------------------------------------------------------------|--------------------------------|
| Founders            | Daily             | Burn rate, runway, active users, revenue                                  | Executive summary + alerts     |
| Investors           | Monthly/Quarterly | LTV:CAC, MRR/ARR, churn, growth rate                                      | Trend charts + cohort analysis |
| Operations managers | Real-time / Daily | Doctor utilization, response time, ticket volume, consultation completion | Operational dashboard          |

### 6.2 Pre-Pilot Dashboard

| Panel                | Metrics Shown                                         | Visualization        |
|----------------------|-------------------------------------------------------|----------------------|
| Build Progress       | Feature completion %, bugs open/closed, test coverage | Progress bars        |
| Internal Usage       | DAU/MAU, sessions per user, feature adoption heatmap  | Line chart + heatmap |
| Burn Tracker         | Monthly spend vs budget, runway remaining             | Countdown gauge      |
| Consultation Testing | Completion rate, average duration, failure modes      | Bar chart            |

### 6.3 Pilot 1 Dashboard

| Panel          | Metrics Shown                                                   | Visualization                |
|----------------|-----------------------------------------------------------------|------------------------------|
| User Growth    | Total registered, MAU, DAU, DAU/MAU ratio                       | Line chart with growth rate  |
| Revenue        | Total revenue, ARPU, transactions by type                       | Stacked bar chart            |
| Unit Economics | CAC, estimated LTV, LTV:CAC trajectory                          | Trend line with target bands |
| Burn & Runway  | Monthly burn rate, cash remaining, runway months                | Gauge + forecast line        |
| Operations     | Doctor utilization, consultation completion rate, response time | Real-time gauges             |
| Quality        | NPS (if collected), support ticket volume, resolution time      | Scorecard                    |

### 6.4 Pilot 2 Dashboard

| Panel              | Metrics Shown                                                                  | Visualization             |
|--------------------|--------------------------------------------------------------------------------|---------------------------|
| Growth & Retention | MAU growth rate, churn rate by cohort, reactivation rate                       | Cohort retention curves   |
| Revenue            | MRR, ARR, revenue by stream, ARPU by segment                                   | Revenue waterfall + trend |
| Unit Economics     | CAC by channel, LTV by cohort, LTV:CAC ratio, payback period                   | Scatter plot + trend      |
| Burn & Runway      | Burn rate trend, cash position, months to breakeven forecast                   | Dual-axis chart           |
| Operational Health | Doctor utilization, coverage hours, callback conversion, admin resolution time | Multi-gauge dashboard     |
| Market Metrics     | Users by geography, facility partners onboarded, pharmacy/lab network size     | Map + counters            |
| Quality            | NPS, CSAT, support ticket trend, consultation completion rate                  | Trend lines               |

### 6.5 Production Dashboard

All Pilot 2 panels plus:

| Panel         | Metrics Shown                                                         | Visualization                            |
|---------------|-----------------------------------------------------------------------|------------------------------------------|
| P&L Summary   | Revenue, COGS, gross margin, operating expenses, net margin           | Monthly P&L waterfall                    |
| Investor View | ARR, ARR growth rate, LTV:CAC, burn multiple, months to profitability | KPI tiles with trend arrows              |
| Predictive    | Revenue forecast (3-month), churn prediction, CAC efficiency forecast | Forecast lines with confidence intervals |
| Compliance    | Uptime %, data breach incidents, regulatory milestone tracker         | Status indicators                        |

### 6.6 Alert Thresholds

| Alert           | Trigger                                       | Severity | Action                                           |
|-----------------|-----------------------------------------------|----------|--------------------------------------------------|
| Runway critical | < 2 months remaining                          | CRITICAL | Trigger contingency plan (see Document 10)       |
| Runway warning  | < 4 months remaining                          | WARNING  | Begin fundraising preparation or scope reduction |
| Churn spike     | Monthly churn increases > 5 percentage points | WARNING  | Root cause analysis within 48 hours              |



|                  |                                              |          |                                                  |
|------------------|----------------------------------------------|----------|--------------------------------------------------|
| Utilization low  | Doctor utilization < 30% for 2+ weeks        | WARNING  | Reduce doctor hours or increase marketing        |
| Utilization high | Doctor utilization > 85% for 1+ week         | WARNING  | Add doctor shifts; patient wait times at risk    |
| Completion drop  | Consultation completion rate drops below 70% | CRITICAL | Technical investigation + ops review immediately |
| NPS decline      | NPS drops > 15 points quarter-over-quarter   | WARNING  | User research and service review                 |

## 7. Benchmarks

### 7.1 East African Health Tech Benchmarks

These benchmarks are derived from publicly available data on comparable companies operating in East Africa (M-TIBA, Babyl/Babylon Health Rwanda, MyDawa Kenya, mPharma, Ilara Health) as of 2025-2026.

| Metric                  | East African Health Tech Range | Health Hub Target (Production) | Assessment                                                                         |
|-------------------------|--------------------------------|--------------------------------|------------------------------------------------------------------------------------|
| CAC                     | \$3-\$10                       | < \$2                          | Aggressive but achievable via partner clinic channel and admin-assisted onboarding |
| LTV                     | \$15-\$50                      | \$30-\$60                      | Within range; higher end dependent on multi-service usage per user                 |
| LTV:CAC                 | 1.5x-4x                        | > 3x                           | Aligned with regional best performers                                              |
| Monthly churn           | 8-20%                          | < 8%                           | Ambitious; requires strong retention loops (prescriptions, follow-ups, callbacks)  |
| ARPU                    | \$1-\$5/mo                     | \$5-\$15/mo                    | Higher than regional average; justified by multi-service platform + premium tier   |
| Gross margin            | 30-55%                         | 40-60%                         | Competitive; platform model should yield higher margins than clinic-heavy models   |
| Doctor utilization      | 35-55%                         | > 60%                          | Above regional average; requires excellent scheduling and demand management        |
| DAU/MAU                 | 10-20%                         | > 25%                          | Significantly above regional average; requires strong daily engagement hooks       |
| NPS                     | 25-45                          | > 50                           | Ambitious; top-quartile for healthcare in emerging markets                         |
| Consultation completion | 65-80%                         | > 85%                          | Above regional average; enabled by admin-assisted model and employed doctor pool   |

### 7.2 Global Telehealth Benchmarks

These benchmarks are derived from publicly available data on global telehealth platforms (Teladoc, Amwell, Doctor on Demand, Practo, KRY/Livi) as of 2025-2026.

| Metric                  | Global Telehealth Range | Health Hub Target (Production) | Assessment                                                                                   |
|-------------------------|-------------------------|--------------------------------|----------------------------------------------------------------------------------------------|
| CAC                     | \$25-\$150              | < \$2                          | Dramatically lower; reflects emerging market dynamics and organic/partner-driven acquisition |
| LTV                     | \$200-\$1,500           | \$30-\$60                      | Much lower in absolute terms; reflects lower pricing corridors in East Africa                |
| LTV:CAC                 | 3x-8x                   | > 3x                           | Within global healthy range                                                                  |
| Monthly churn           | 3-8%                    | < 8%                           | Higher churn acceptable given lower switching costs in emerging markets                      |
| ARPU                    | \$15-\$80/mo            | \$5-\$15/mo                    | Lower ARPU offset by lower delivery costs (Indian ops staff per params.md Section 4.1)       |
| Gross margin            | 40-65%                  | 40-60%                         | Comparable; technology-platform economics are geography-neutral                              |
| Doctor utilization      | 50-70%                  | > 60%                          | Within global healthy range                                                                  |
| DAU/MAU                 | 15-30%                  | > 25%                          | Within global healthy range                                                                  |
| NPS                     | 40-70                   | > 50                           | Within global healthy range                                                                  |
| Consultation completion | 80-92%                  | > 85%                          | Within global healthy range                                                                  |

### 7.3 Benchmark Positioning Summary

| Dimension        | Health Hub vs East African Peers | Health Hub vs Global Telehealth      |
|------------------|----------------------------------|--------------------------------------|
| Acquisition cost | Lower CAC (partner-driven)       | Dramatically lower (100x less)       |
| Monetization     | Higher ARPU (multi-service)      | Lower ARPU (emerging market pricing) |
| Retention        | More ambitious churn targets     | Comparable churn expectations        |
| Efficiency       | Higher utilization targets       | Comparable utilization               |
| Satisfaction     | Higher NPS targets               | Comparable NPS                       |
| Margins          | Comparable gross margins         | Comparable gross margins             |

|                     |                                |                                                   |
|---------------------|--------------------------------|---------------------------------------------------|
| Overall positioning | Premium operator in the region | Emerging market adaptation of proven global model |
|---------------------|--------------------------------|---------------------------------------------------|

**Key insight:** Health Hub's competitive advantage is the combination of low CAC (partner-driven, admin-assisted acquisition), multi-service ARPU (not just consultations but pharmacy + diagnostics + care coordination), and low delivery costs (India-based ops staff). This triangulation creates unit economics that are competitive with global benchmarks despite dramatically lower absolute pricing.

## 8. Cross-References

| Topic                                       | Document                                             |
|---------------------------------------------|------------------------------------------------------|
| Foundation parameters and assumptions       | <a href="#">params.md</a>                            |
| Cost breakdown by phase and scenario        | Document 03 — Cost Breakdown                         |
| Revenue modelling across 21 combinations    | Document 05 — Revenue Modelling                      |
| Pricing strategy and corridors              | Document 06 — Pricing Strategy                       |
| Customer acquisition strategy and channels  | Document 08 — Customer Acquisition                   |
| Cash flow and runway analysis               | <a href="#">Document 10 — Cash Flow &amp; Runway</a> |
| Competitive analysis and market positioning | Document 15 — Competitive Analysis                   |

End of Document 09 — Business Metrics. All figures trace to params.md v2.